

Instruction	op	funct3	Funct7	Type
add	0110011 (51)	000 (0)	0000000 (0)	R-Type
sub	0110011 (51)	000 (0)	0100000 (32)	R-Type
and	0110011 (51)	111 (7)	0000000 (0)	R-Type
or	0110011 (51)	110 (6)	0000000 (0)	R-Type
addi	0010011 (19)	000 (0)	-	I-Type
beq	1100011 (99)	000 (0)	-	B-Type
bne	1100011 (99)	001 (1)	-	B-Type
lw	0000011 (3)	010 (2)	-	I-Type
sw	0100011 (35)	010 (2)	-	S-Type
jal	1101111 (111)	-	-	J-Type
jalr	1100111 (103)	000 (0)	-	I-Type
lui	0110111 (55)	-	-	U-Type

Instruction Bits

funct7							4 3 2 1 0					rs1				funct3				rd				
11 10 9 8 7 6 5							4 3 2 1 0					rs1				funct3				rd				
11 10 9 8 7 6 5							rs2					rs1				funct3				4 3 2 1 0				
12 10 9 8 7 6 5							rs2					rs1				funct3				4 3 2 1 11				
31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12																rd								
20 10 9 8 7 6 5 4 3 2 1 11 19 18 17 16 15 14 13 12																rd								
31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7																								

instruction bit	31	31	31	31	30:25	24:21	20	I S B U J
	31	31	31	31	30:25	11:8	7	
	31	31	31	30	29:25, 11	10:7	0	
	31	30:20	19:12	0	0	0	0	
	31	31	19:12	20	21:16	15:12	0	
31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0								

Immediate Bits

Name	Register Number	Usage
zero	x0	Constant value 0
ra	x1	Return address
sp	x2	Stack pointer
gp	x3	Global pointer
tp	x4	Thread pointer
t0-2	x5-7	Temporaries
s0/fp	x8	Saved register / Frame pointer
s1	x9	Saved register
a0-1	x10-11	Function arguments / return values
a2-7	x12-17	Function arguments
s2-11	x18-27	Saved registers
t3-6	x28-31	Temporaries

7 bits	5 bits	5 bits	3 bits	5 bits	7 bits
funct7	rs2	rs1	funct3	rd	op
imm _{11:0}		rs1	funct3	rd	op
imm _{11:5}	rs2	rs1	funct3	imm _{4:0}	op
imm _{12,10:5}	rs2	rs1	funct3	imm _{4:1,11}	op
imm _{31:12}				rd	op
imm _{20,10:1,11,19:12}				rd	op
20 bits				5 bits	7 bits

R-Type
I-Type
S-Type
B-Type
U-Type
J-Type